Math 4360H: Assignment III

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Problem 1. Find all ideals of the ring $\mathbb{Z}_{60}$. Identify the prime ideals and the maximal ideals (justify).

Solution 1. We defined an ideal to be an additive subgroup. Since $(\mathbb{Z}_{60}, +)$ is cyclic we know that there is a subgroup for each divisor of 60. These divisors are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, and 60. The subgroups $\langle 1 \rangle$ and $\langle 60 \rangle$ are obviously ideals, the rest must be checked. Recall that an additive subgroup is an ideal I if (in the commutative case) for $i \in I$ and $r \in R$, $ri \in I$. This is clearly the case for every divisor we found as ri will just be kn for k a divisor and $n \in \mathbb{Z}$, $0 \leq n < 60$ and kn is hence an integer multiple of the divisor and will therefore belong to the additive subgroup $\langle k \rangle$ which is the ideal. I will check the maximal ideals first as they will also be prime. Recall that a maximal ideal of a ring R is an ideal which is not properly contained in any other ideal of R. Borrowing from normal subgroups, for generators a and b, $\langle a \rangle \subset \langle b \rangle$ iff $a \in \langle b \rangle$. Clearly then the only maximal ideals will be those with prime generators because $\langle p \rangle$ will only contain things like np for $n \in \mathbb{Z}$. The only prime-generated ideals we found were $\langle 2 \rangle$, $\langle 3 \rangle$, and $\langle 5 \rangle$ and so these are the only maximal ideals. We get for free the previously mentioned maximal ideals as also being prime ideals. We can quickly show that others are not prime as

(i) $2 \cdot 2 = 4 \in \langle 4 \rangle$, $2 \notin \langle 4 \rangle$

(ii) $3 \cdot 2 = 6 \in \langle 6 \rangle$, $2 \notin \langle 6 \rangle$

(iii) $5 \cdot 2 = 10 \in \langle 10 \rangle$, $2 \notin \langle 10 \rangle$

(iv) $2 \cdot 6 = 12 \in \langle 12 \rangle$, $2 \notin \langle 12 \rangle$

(v) $3 \cdot 5 = 15 \in \langle 15 \rangle$, $3 \notin \langle 15 \rangle$

(vi) $2 \cdot 10 = 20 \in \langle 20 \rangle$, $2 \notin \langle 20 \rangle$

(vii) $2 \cdot 15 = 30 \in \langle 30 \rangle$, $2 \notin \langle 30 \rangle$

however, $\langle 60 \rangle$ is a prime ideal as it contains only $\{0\}$ and the only way to get zero is to multiply by some multiple of 60, all of which are "in" $\langle 60 \rangle$ when under mod 60.

Problem 2. Let $I = \{(3x, y) \mid x, y \in \mathbb{Z}\}$

(a) Show that I is a maximal ideal of the ring $\mathbb{Z} \times \mathbb{Z}$.

(b) What if $3x$ is replaced with $4x$?

(c) Can we make a general statement about this?

Solution 2.

(a) $\mathbb{Z} \times \mathbb{Z}/I$ is the set of remainders of element-wise division of tuples. Breaking this down across the direct product this looks like $\mathbb{Z}/\{3x \mid x \in \mathbb{Z}\} = \mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/\{y \in \mathbb{Z}\} = \mathbb{Z}/\mathbb{Z}$ which is just $\mathbb{Z}_3 \times \{0\} \cong \mathbb{Z}_3$. Since p is prime, $\mathbb{Z}_3$ is a field which implies the quotienting set I is maximal.

(b) Using the same tactic, $\mathbb{Z} \times \mathbb{Z}/\{(4x, y) \mid x, y \in \mathbb{Z}\} \cong \mathbb{Z}_4 \times \{0\} \cong \mathbb{Z}_4$ which is not a field as 4 is not prime and hence the ideal $\{(4x, y) \mid x, y \in \mathbb{Z}\}$ is not maximal.

(c) Yes, ideals of the form $\{(px, y) \mid x, y \in \mathbb{Z}\}$ for prime $p \in \mathbb{Z}$ are maximal in $\mathbb{Z} \times \mathbb{Z}$. This is because you end up with the $\mathbb{Z}_p$ structure which is a field and hence the ideal was maximal.

Problem 3. Consider the ring $R = \mathbb{Z}_8 \times \mathbb{Z}_{30}$.

(a) Find all maximal ideals of the ring $R = \mathbb{Z}_8 \times \mathbb{Z}_{30}$.

(b) For each maximal ideal I, determine the size of the field R/I.

Solution 3.

(a) Intuitively I want to say that the ideals of this will be the direct product of the ideals of $\mathbb{Z}_8$ and $\mathbb{Z}_{30}$. I don't think we proved this in class so I'll attempt it here:

Proof. Let I be an ideal of the ring $R \times S$. We want to show that the set

$$I_R = \{x \in R \mid (x, y) \in I, y \in S\}$$

is an ideal of R (and the same for $I_S = \{y \in S \mid (x, y) \in I, x \in R\}$ being an ideal of S). For some $r \in R$ and $(x, y) \in I$

$$(r, 0_S) \cdot (x, y) = (rx, 0_S) \in I$$

because I is an ideal. I_R is evidently an ideal of R with the second element just tacked on. The same can be shown for I_S.

Now that we have a definition for ideals of R and S which work, we want to show that $I = I_R \times I_S$.

(a) Clearly $I \subseteq I_R \times I_S$ as for some $(x, y) \in I$, $x \in I_R$ and $y \in I_S$.

(b) For $I \supseteq I_R \times I_S$ consider an $(x, y) \in I_R \times I_S$. Then since $x \in I_R$ there is a $y' \in I_S$ such that $(x, y') \in I$ as we defined I_R that way. We can also find, for $y \in I_S$ an $x' \in I_R$ so that $(x', y) \in I$. Now if R and S have a multiplicative identity (which in our case they do) we can take $(1_R, 0_S) \cdot (x, y') = (x, 0_S) \in I$ since I is an ideal. We can do the same for the x' element, $(0_R, 1_S) \cdot (x', y) = (0_R, y) \in I$ again because I an ideal. If we sum these results we obtain that $(x, y) \in I$, i.e. all things in $I_R \times I_S$ are in I so $I \supseteq I_R \times I_S$. $\square$

I believe this is sufficient as a proof but I also think I may have overthought things after some discussion with classmates. I'd be interested to see if there is an easier proof as I can't find anything online (I only looked it up after doing this) that uses techniques I understand.

Now we know that the maximal ideals of $\mathbb{Z}_k$ are the sets $\langle n \rangle$ for which n is one of the prime factors of k. Hence the only proper maximal ideal of $\mathbb{Z}_8$ is $\langle 2 \rangle$ and the proper maximal ideals of $\mathbb{Z}_{30}$ are $\langle 2 \rangle$, $\langle 3 \rangle$, and $\langle 5 \rangle$. If we want something that is maximal in the direct product then the quotient would have to give us a field. This necessitates that one of the parts of the direct product ideal, $I = I_R \times I_S$ is the entirety of one of the rings $(R \times S)$ being quotiented else we could take elements $(r, 0) \in R$ and $(0, s) \in S$ and multiply their cosets in $R \times S/I$ as $((r, 0) + I)((0, s) + I) = 0 + I$ which means $((r, 0) + I), ((0, s) + I)$ would be zero divisors. If one of I_R or I_S is the entirety of R or S then we would end up quotienting out the whole of either R or S and end up with things that act like tuples with one entry always zero, hence giving no zero divisors. This means that the maximal ideals of $\mathbb{Z}_8 \times \mathbb{Z}_{30}$ can only be $\langle 2 \rangle \times \mathbb{Z}_{30}$, $\mathbb{Z}_8 \times \langle 2 \rangle, \langle 3 \rangle, \langle 5 \rangle$.

(b) (i) $\mathbb{Z}_8 \times \mathbb{Z}_{30}/\langle 2 \rangle \times \mathbb{Z}_{30}$: this will look like $\{(a, b) + \langle 2 \rangle \times \mathbb{Z}_{30} \mid a \in \mathbb{Z}_8, b \in \mathbb{Z}_{30}\}$ which I believe is the direct product $\{a + \langle 2 \rangle \mid a \in \mathbb{Z}_8\} \times \{b + \mathbb{Z}_{30} \mid b \in \mathbb{Z}_{30}\}$ which is a field (because the ideal is maximal) of two elements times a field of one element and is hence isomorphic to $\mathbb{Z}_2$.

(ii) $\mathbb{Z}_8 \times \mathbb{Z}_{30}/\mathbb{Z}_8 \times \langle 2 \rangle$: As above this is $(\mathbb{Z}_8/\mathbb{Z}_8) \times (\mathbb{Z}_{30}/\langle 2 \rangle) \cong 0 \times \{a + \langle 2 \rangle \mid a \in \mathbb{Z}_{30}\}$. We can see that $\{a + \langle 2 \rangle \mid a \in \mathbb{Z}_{30}\}$ will have only two elements, repeated for odd and even a. This is again the two element structure $\mathbb{Z}_2$

(iii) $\mathbb{Z}_8 \times \mathbb{Z}_{30}/\mathbb{Z}_8 \times \langle 3 \rangle$: As previous we can break this down quickly to $\{a + \langle 3 \rangle \mid a \in \mathbb{Z}_{30}\}$ which will only have 3 unique elements, one which contains 0, one which contains 1 and one which contains 2. Since this is a field it is isomorphic to $\mathbb{Z}_3$

(iv) $\mathbb{Z}_8 \times \mathbb{Z}_{30}/\mathbb{Z}_8 \times \langle 5 \rangle$: As previous this is isomorphic to $\mathbb{Z}_5$ and hence has 5 elements.

Problem 4. Show by example that the intersection of two prime ideals need not be a prime ideal.

Solution 4. The ring $\mathbb{Z}$ has prime ideals $\langle 2 \rangle$ and $\langle 3 \rangle$ since $\langle 2 \rangle = 2\mathbb{Z}$ and clearly any $ab \in 2\mathbb{Z}$ requires that either a or b is a multiple of two and is hence in $2\mathbb{Z}$. The same can be argued for $\langle 3 \rangle$. Their intersection is all things that are multiples of 3 *and* 2, i.e. multiples of $2 \cdot 3 = 6$ so $\langle 6 \rangle = \langle 2 \rangle \cap \langle 3 \rangle$. $\langle 6 \rangle$ is not prime as $2 \cdot 3 = 6 \in \langle 6 \rangle$ but neither 3 nor 2 are in $\langle 6 \rangle$.

Problem 5. Let R be a principal ideal domain and I an ideal of R. Show that every ideal of R/I is principal.

Solution 5. I'm not happy with this but it's the best I have: We know that the map $\phi : R \to R/I$ given by $\phi(a) = a + I$ is a homomorphism. We also know that (in the commutative case) and ideal of R is a set where for $r \in R$ and $i \in I$, $ri \in I$. Consider an ideal J of R. Because J is an ideal we have that for $j \in J$ and $r \in R$ $rj \in J \implies \phi(rj) \in \phi[J]$ so $\phi(r)\phi(j) \in \phi[J]$. This shows that $J + I$, i.e. $\{j + I \mid j \in J\} = J/I$ is an ideal of R/I because $\phi(r)\phi(j) \in \phi[J]$ satisfies the definition of an ideal for elements of the form $\phi(j) = j + I$ and $\phi(r) \in R/I$. Since we chose J as an ideal of R it is principal as R is a PID and since J is arbitrary, all ideals of R/I are principal.

Problem 6. Let $R = \{a + bi \mid a, b \in \mathbb{Z}\}$. Show that $R/\langle 1 - i\rangle$ is a field, and determine its number of elements.

Solution 6. In $R/\langle 1 - i\rangle$ we can treat i as x, an indeterminate. Then this becomes a polynomial ring quotiented by a polynomial ideal. Specifically it is all integer coefficient polynomials of degree at most 1 quotiented by the ideal generated by a polynomial of degree 1. Since here, with $I = \langle 1 - i\rangle$, $1 - i + I = I \equiv 0$ and so, working "mod $1 - i$", $1 - i \equiv 0 \implies 1 \equiv i$ however we define i to be the number where $i^2 = -1$ so $1^2 \equiv -1 \implies 2 \equiv 0$. At this point this feels like $\mathbb{Z}_2$ but let's check. For a general $a + bi \in R$ we've determined that, mod $1 - i$, $a + bi \equiv a + b$. And we've shown that if $a + b$ contains a factor of 2, $a + b \equiv 0$. If $a + b$ does not contain a factor of 2 we've shown that $a + b \equiv 1$. This is in fact $\mathbb{Z}_2$ which we know to be a field with two elements.

Figure 1: Space Filling Cat Picture. Schnitzel M. at left, Biscuit G. at right.